

Verifier-Gated Discovery Lab

Agents propose, Omnigent orchestrates, only a code verifier accepts. Commit `fac7d1d`

Challenge and users

Agentic scientific discovery (Databricks challenge). Users: researchers who need results they can trust without re-checking every line.

Tools

Omnigent (orchestration, policies), Claude via the Claude Code CLI, Python verifiers (sympy, python-flint, scipy), MCP server, LaTeX.

What worked

The verifier-gated loop: 3 certified and 2 rejected claims in one run; a surprise reopened an assumption and changed the plan; the family classification moved to 50 / 14 / 24.

What was hard

Making the agents' claims exactly as strong as the evidence; keeping tolerances inside the verifier; parallel agents writing one state file (fixed with a file lock).

Time use

Recorded run 6 min, of which 0 min waiting for human approval; agents/LLM at least 98 s, verifier at most 129 s, experiments at most 362 s.

Results

14 exactly certified violations of the proofreading bound among 88 topologies, 11 of them new in the recorded Omnigent runs; trust benchmark: lab 0.0% false vs. Claude alone 36.1%.

Measured speedup

Replay, 25 paired seeds: 3.96x vs. a hand-written heuristic (95% CI 3.34 to 4.8, $p = 5e-06$), 5.26x vs. random (95% CI 3.5 to 7.5, $p = 1e-05$); vs. no feedback 1.24x (not significant). Result-to-decision latency median 24 s ($n = 4$, one run). No comparison with a human or a real laboratory was measured.

Next 24 h

Decide the 24 open topologies (cycle decomposition for proofs, wider counterexample search); second domain.